Luca Atella

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Summary

Software Architect, Tech Lead, and Backend/Platform Engineer focused on **modular systems**, **cloud-portable architectures**, **runtime governance**, and **long-lived backend platforms**.

I design and directly build backends, APIs, and application infrastructure. At Vemar, from April 2021 to October 2024, I worked as Tech Lead with continuous technical coordination responsibility for a 4-person team, personally designed and implemented the infrastructure for 80 trackers, and developed the framework on which delis.app is based.

Certifications

- AWS Certified Solutions Architect - Associate
- AWS Certified Cloud Practitioner
- AWS Certified AI Practitioner
- Cisco Networking Academy - CCNA path

Core Strengths

- Backend and platform architecture for integration-heavy systems
- AWS serverless deployment and managed-service architectures
- Cloud-portable application design and infrastructure abstraction
- Modular systems, plugin-based architectures, and runtime governance
- Docker / Docker Compose local environments and deployment automation
- IoT, edge/cloud integration, and Digital Twin architecture concepts
- Geospatial data processing and 3D terrain model generation

Professional Experience

Tech Lead / Backend & Platform Engineer — Vemar S.A.S.

April 2021 - October 2024

- Continuously coordinated a 4-person technical team
- Developed the plugin-based backend framework on which delis.app is based
- Personally designed and implemented the infrastructure for 80 trackers
- Worked on socket communication, CAN bus data, persistence, and REST API exposure
- Centralized application logic, authentication, API exposure, and persistence concerns
- Defined REST API contracts and backend structure
- Designed plugin loading and execution boundaries
- Introduced Docker-based development and deployment workflows

Selected Projects

Cloud-Portable Fleet Management Platform

- Fleet Management System used by Studio Lambda to manage vehicle turnover for 50 employees
- Built a FastAPI backend and worked on React/TypeScript frontend integration
- Abstracted persistence and object storage through repository and storage contracts
- Used Docker Compose, DynamoDB Local, and MinIO for local development
- Deployed to AWS using Lambda, ECR, DynamoDB, S3, CloudFront, CloudFormation, and CloudWatch

Real-Time Fleet Telemetry Backend

- Infrastructure distinct from the Studio Lambda platform
- Personally designed and implemented at Vemar for 80 trackers
- Included socket communication, CAN bus data, persistence, and REST API exposure

Unified Backend for Heterogeneous Data Sources

- Plugin-based backend framework I developed at Vemar and on which delis.app is based
- Integration separation, explicit contracts, and unified API model presented in SAFE form
- Does not attribute every architectural generalization to delis.app itself

ImportSpy — Policy-as-Code Admission for Python

- Created and maintain an open-source admission layer that evaluates Python code and dependencies before explicit execution
- Static preflight, dependency identity analysis, and evidence/policy evaluation with structured admission decisions
- 15,000+ PyPI downloads across project releases
- Talk on explicit module contracts at GDG Basilicata, 20 May 2025

B3DO — Basilicata 3D Open

- Unpublished geospatial processing pipeline for transforming public Basilicata terrain datasets into 3D terrain models
- CLI-driven workflow for DTM merging, boundary extraction, raster clipping, resampling, LOD mesh generation, and textured previews
- Uses GDAL, Rasterio, NumPy, PyVista, Fiona, and Typer
- Separates raw data, processed datasets, build artifacts, and model outputs for reproducible data-to-model processing

IoT Data Aggregation Architecture for Digital Twin Systems

- Conceptual edge-to-cloud architecture for heterogeneous IoT data acquisition and aggregation
- Includes BLE/Zigbee mesh, gateway ingestion, GraphQL API, data-specific layers, microfrontends, storage gateway, and AI-assisted processing

Runtime Compatibility for Agenzia delle Entrate Applications

- Professional intervention in an accounting firm Windows environment
- Isolated Java runtimes for IRAP 2019-2021, ISA, Desktop Telematico / Entratel, and other tools
- Handled JNLP/LaunchAnywhere startup modes and Entratel TLS/truststore requirements
- Infrastructure and compatibility work, not tax consulting or regulatory certification

Education

BSc in Computer Science and Technology (L-31) — University of Basilicata
2018 - Present

High School Diploma in Electronics, Electrical Engineering, and Automation
2013 - 2018

Languages

Italian: native · English: B1, actively improving

Tech

Python · FastAPI · React · AWS · Lambda · DynamoDB · S3 · CloudFront · CloudFormation · Docker · Docker Compose · Kubernetes · Linux · Networking · Runtime Governance · GDAL · Rasterio · PyVista